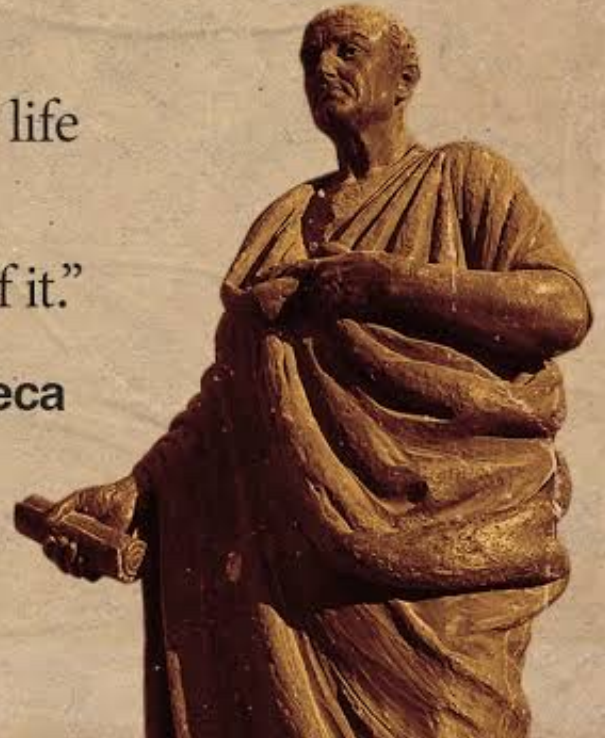


“It is not that we are given a short life
but we make it short, and we
are not ill-supplied but wasteful of it.”

- Seneca

StoicQuotes.com



Agenda

- ① Review Apache Airflow
- ② Demo for Airflow
- ③ Real life example

AIRFLOW

- ① Introduction to Apache Flink
- ② How does flink work internally
- ③ Spark vs Flink

FLINK

Request: Pager be away for 2 mins → 11:15 pm

Review Apache Airflow

Open source platform for developing, scheduling and monitoring

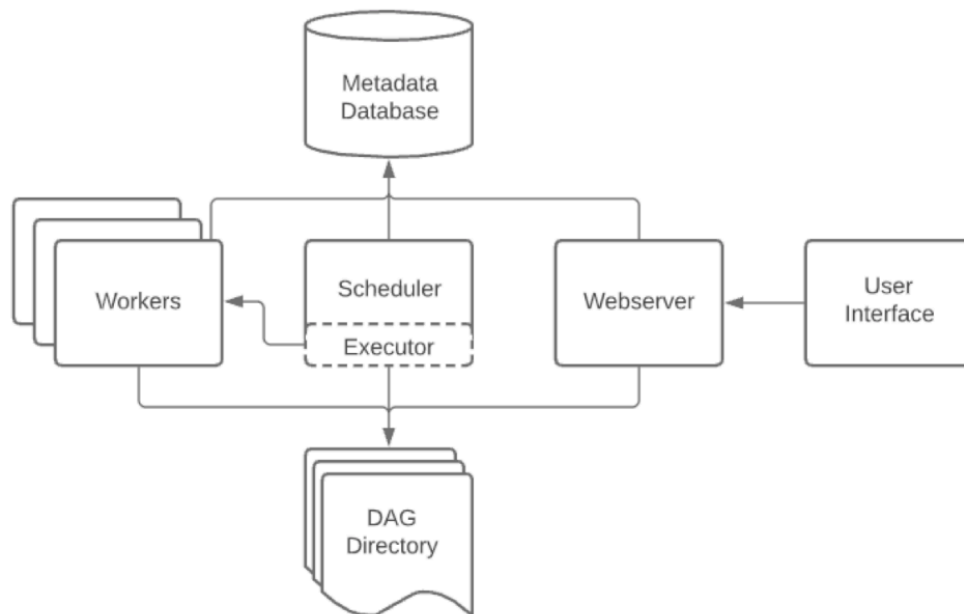
batch oriented workflows

- ① Designed for orchestrating complex workflows
- ② Built using python, you can define workflows or DAGs as code
- ③ Open source - primarily developed by Airbnb

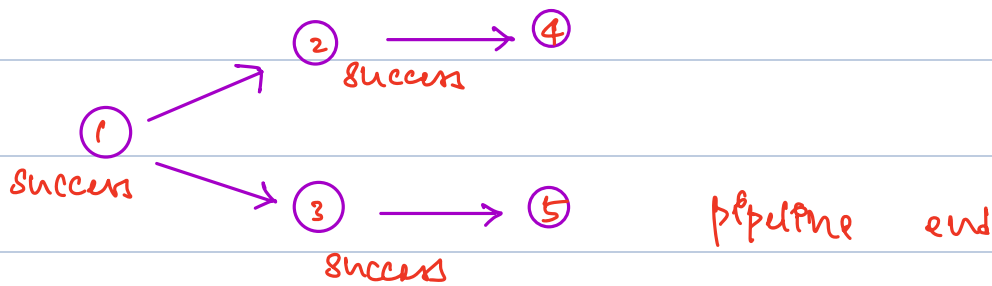
GCP - composer

AWS - Managed Workflows

Airflow Components



DAG - Heart of your Airflow



Extract Transform

load

dag 1

task-1-dag.py

scheduler - Background job that monitors DAG folder, metadata and triggers workers

Metadata Database - Store details about tasks & run

Webserver & UI - User to interact & monitor different Jobs in our workflow

workers - VM's where Jobs are run

Possible DAG structure

Data Gathering - Structured / Semi-Structured / Unstructured

Data Ingestion - HDFS / AWS

Data Preprocessing -

ETL

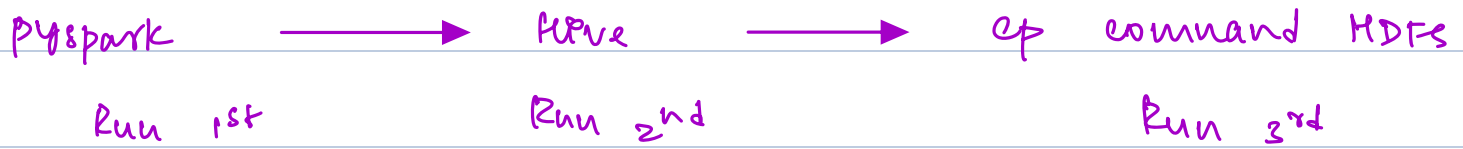
Visualization

Hugging face

- AI models [Open sourced]

[1st down new model → speech to text]

Real life Alerts Example



Demo 01 and Demo 02

01. Basic DAG creation
02. Real world - current Project

Apache Flink

Open source distributed data processing framework used for handling both streaming and batch pipelines

real time streaming

Features of Apache Flink

- ① Event driven Architecture
- ② Stateful Processing
- ③ Fault tolerant
- ④ Scalable
- ⑤ True Stream Processing ms, ms lat
- ⑥ Event time Processing

Bank transactions

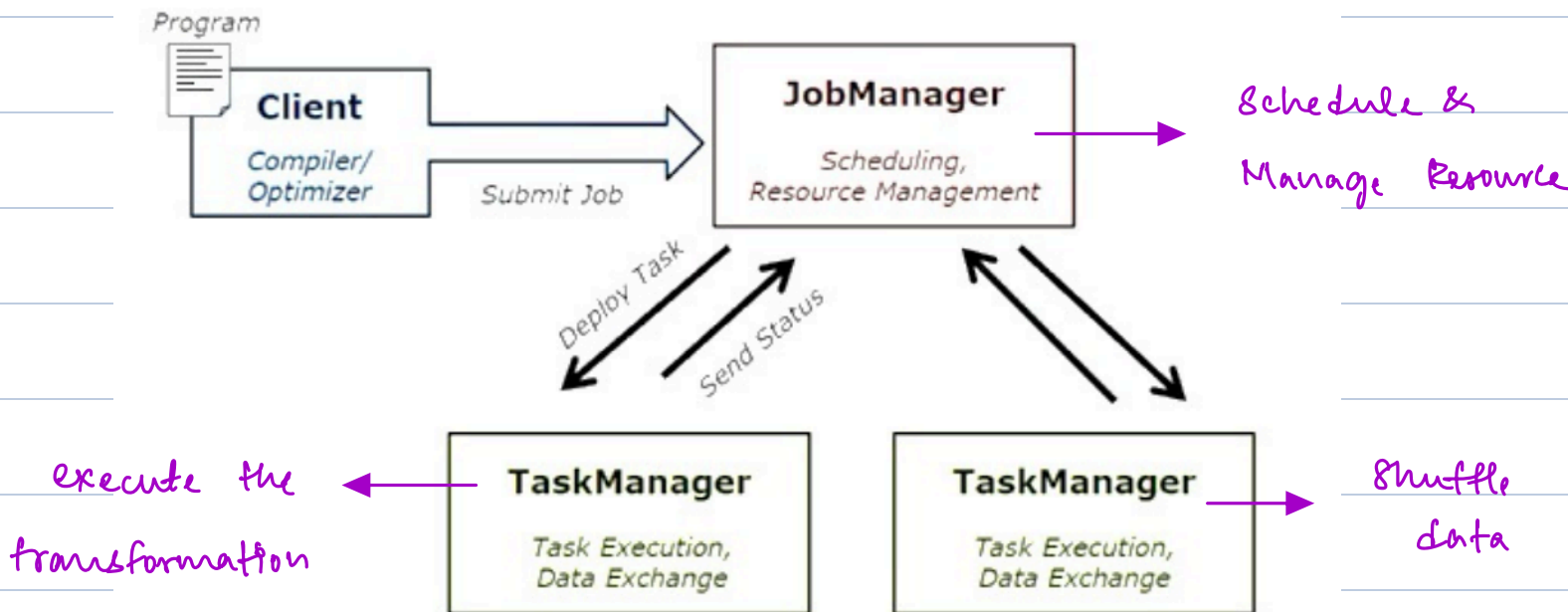
Gautam, 50	Annu, 20
Dheeraj, 5	abhir, 40

Spark stream

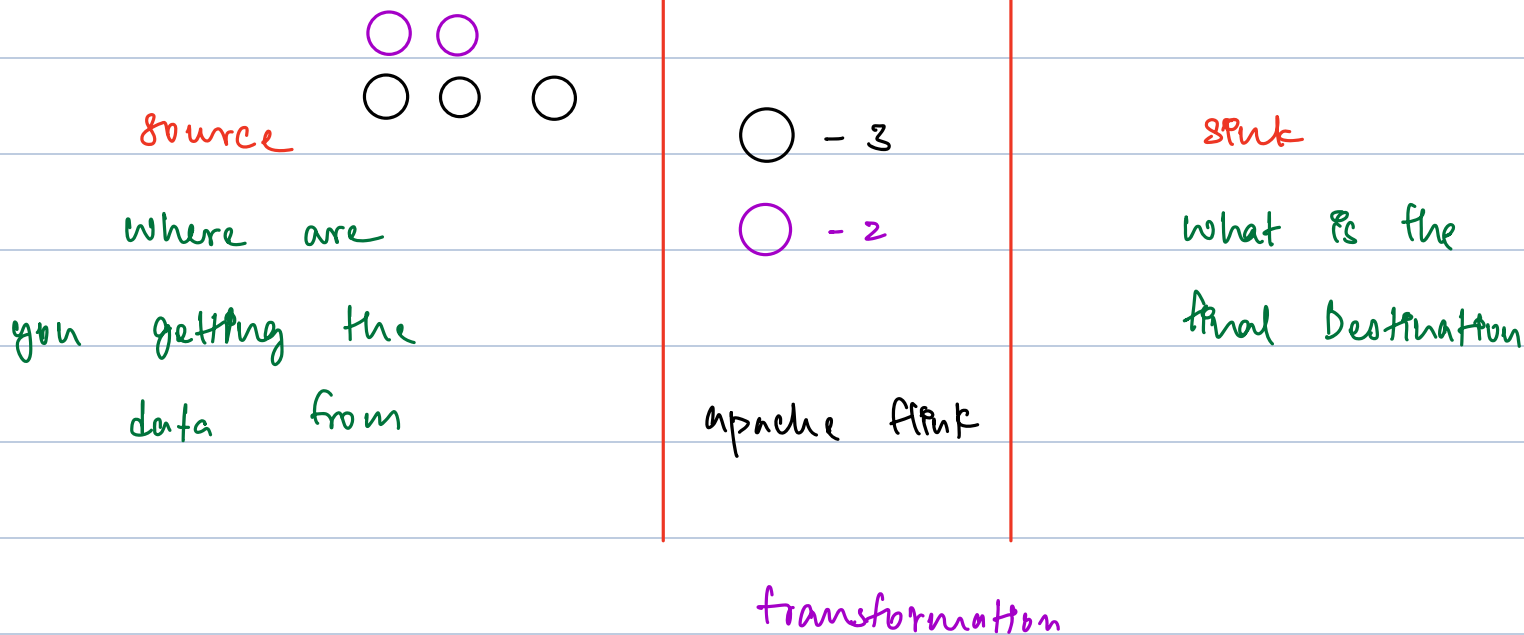
4 transaction are treated
in 1 micro batch

Basic Architecture

[master slave architecture]



Streaming



Apache spark

- ① Both batch and micro batch Ps handled by spark
- ② Introduces latency in spark streaming - near real time
- ③ Window and time limited capabilities with window fn
- ④ Achieve fault tolerance using state and checkpoints
- ⑤ High MC capability

Apache Flink

- ① Optimized for realtime data, can perform batch processing.
- ② Real time stream processing - micro sec
- ③ Session based window functions are available
- ④ Achieve fault tolerance using state and checkpoint
- ⑤ Lesser support for MC at full point in time

Demo



word count